

Impact of Extras and Top Individual Scores on Team Totals in Asia Cup Cricket*

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This paper investigates whether extra runs scored due to bowling errors meaningfully impact a team's total score in cricket matches. Applying simple linear regression, the analysis finds that extras show only a weak correlation with total runs scored. In search of a stronger predictor, we also examine the effect of a team's highest individual batting score using the same approach. Results demonstrate that while extras offer a modest contribution, the highest individual score is strongly associated with total team runs. These findings highlight that star player performances are much more influential than extras in boosting team outcomes, as detailed in the sections that follow.

1 Introduction

The result of a cricket match depends on a lot of factors, such as extras scored, standout individual performances to name a few. In tournaments with high stakes like Asia Cup, it's crucial to understand what drives a team's success. Scoring is one of the crucial aspect for coaches, players and analysts alike. The central question of this paper is: How do extras scored and a team's highest individual score in a match affect the total score of the team? **Extras are runs awarded to the batting team as a result of rule infringements or errors committed by the bowling team.**

This is a significant issue, as the team strategies would generally involve reducing the extras, and experts often highlight the impact of an extraordinary performance of a single player. Existing research focus on statistical relations alone, but main concentration is on the general predictors of match outcomes (Jhanwar and Pudi 2016) and (Kampakis and Thomas 2015)

*Project repository available at: <https://github.com/SaipranavSripathi/Math261-LinearRegression-AsiaCupCricketAnalysis>.

or on players averages (Dalal et al. 2024), rather on the effects of extras and individual performances on team totals.

In this analysis, we analyse match-level data of the Asia Cup from 1984 to 2022, and apply simple linear regression to regress total score with extras scored and highest individual score separately. We discover that extras are weakly related to the total team score, whereas the highest individual score has a major impact on their total scores.

The remainder of this paper is as follows. Section 2 focuses on describing the Asia Cup dataset and its principal variables. Section 3 outlines the methods used for analyzing the effects of extras and highest individual scores on team totals. Section 4 presents the main results and interprets their meaning for understanding cricket scoring. Finally, Section 5 discusses the broader strategic significance of these findings.

2 Data and Variables

This analysis uses match-level data collected from the Asia Cup tournaments, accessible at (SCORENetwork 2022) Data Repository. The dataset contains a record for every match played from the inaugural Asia Cup in 1984 up to 2022, now held every alternate year in various Asian host cities.

To address the central research question, we will analyse 3 principal variables, first, total runs scored by each team in a match as primary response variable. Second, extras represent the additional runs awarded due to bowling errors, including wides, no-balls, byes etc. Third, the highest individual batting score from the team in a match acts as the key predictor for standout performances.

The box plots in Figure 1 helps illustrate the distribution of these variables across the dataset. The plot for total runs scored reveals that team scores in the Asia Cup average around 200, with a broad spread ranging from approximately 125 to 300 runs. This displays the typical scoring environment in high-level tournament matches. When examining extras conceded, most teams allow between 8 and 18 runs per match, although the data includes a few exceptional outliers where extras exceeded 30. The box plot for the highest individual score indicates that top batters generally achieve scores clustered between 50 and 90, where several outstanding performances rising above 130 and sometimes crossing 170.

Each match record includes additional information such as team and opponent names, venue, year, toss, outcome, selection after toss, extras given. The dataset contains a record of roughly 250 matches, representing comprehensive sample of teams in Asia Cup and no missing values were found.

While the Asia Cup features some of the most cricket-intensive nations globally—India, Pakistan, Sri Lanka, Bangladesh, Afghanistan—other prominent cricket countries such as Australia, England, and West Indies are not included, as they do not participate in the Asia Cup.

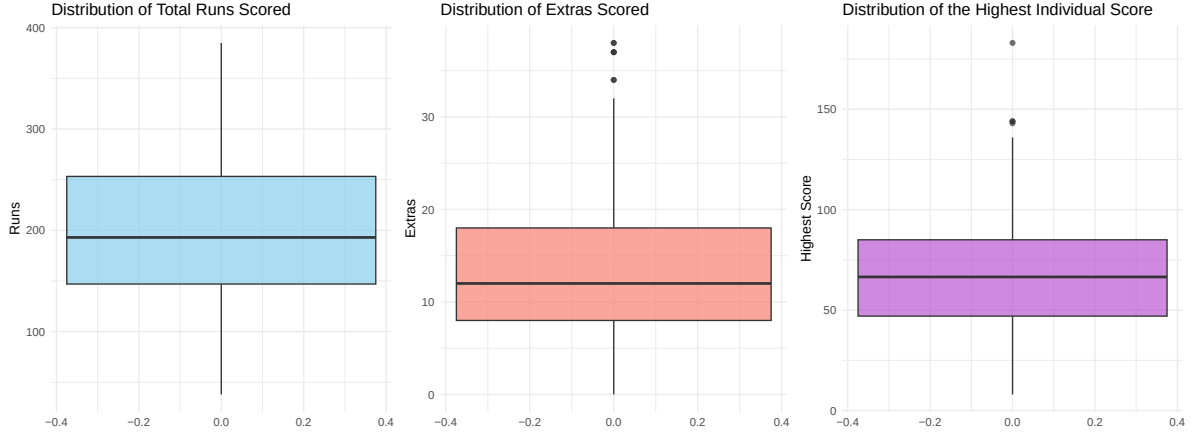


Figure 1: Boxplots showing the distribution of total runs, extras scored, and highest individual score for teams in Asia Cup matches (1984–2022).

This makes generalization difficult: the analysis applies best to Asian teams whose data is being analysed, which comprise the majority of international cricket matches, but may not extend to all cricket-playing nations.

In summary, this dataset is well suited to addressing the question of how extras and highest scores affect team total runs in high-level cricket, though the findings are not fully generalizable to all international cricket contexts.

3 Methods

The affect of extras scored and highest individual batting score on teams total is examined by two separate simple linear regression models to the entire Asia Cup data available.

First Regression: Effect of Extras Scored

Let $RunScored_i$ denote total runs scored by team i , and $ExtrasScored_i$ the number of extras recorded in match i . The model is:

$$RunScored_i = \beta_0 + \beta_1 * ExtrasScored_i + \epsilon_i$$

where β_0 is the intercept, β_1 measures the expected change in total runs per additional extra run, and ϵ_i is the error term for match i which represents the unexplained variation of total runs scored. And we assume that these errors are independent from each other, distributed normally with constant variance.

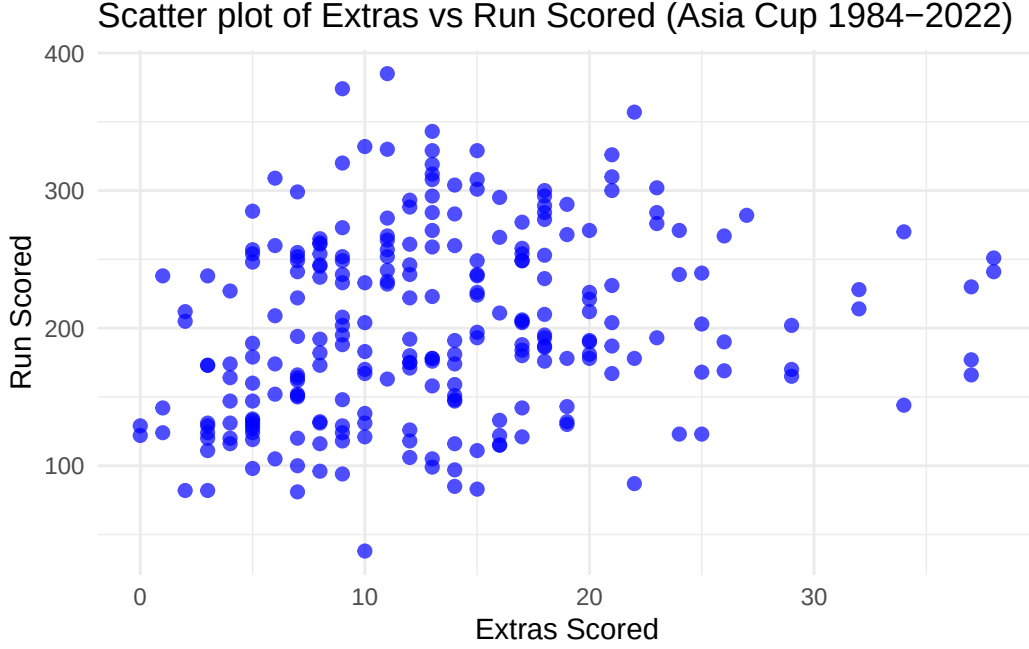


Figure 2: Relation between extras scored and total team runs in Asia Cup matches

The scatter plot in Figure 2 describes of extras scored versus total runs scored across all Asia Cup matches from 1984 to 2022 reveals a modest/low but statistically significant positive relationship. In the visualization, each point represents a team's totals for a given match with the extras scored on corresponding axes.

Second Regression: Effect of Highest Individual Score

Let $HighestScore_i$ be the highest individual run total for team i in match. The model is:

$$RunScored_i = \beta_0 + \beta_1 * HighestScore_i + \epsilon_i$$

Here, β_0 is the intercept, β_1 captures the effect of the top individual score, and ϵ_i is the unexplained variation for match.

The scatter plot in Figure 3 describes of highest individual score versus total runs scored across all Asia Cup matches from 1984 to 2022 reveals a high and significant positive relationship. In the visualization, each point represents a team's totals for a given match with the highest individual score on corresponding axes.

Parameter estimations are done using the `lm()` function in the R programming language (R Core Team 2024) with data visualization and manipulation using ggplot2 and dplyr (H.

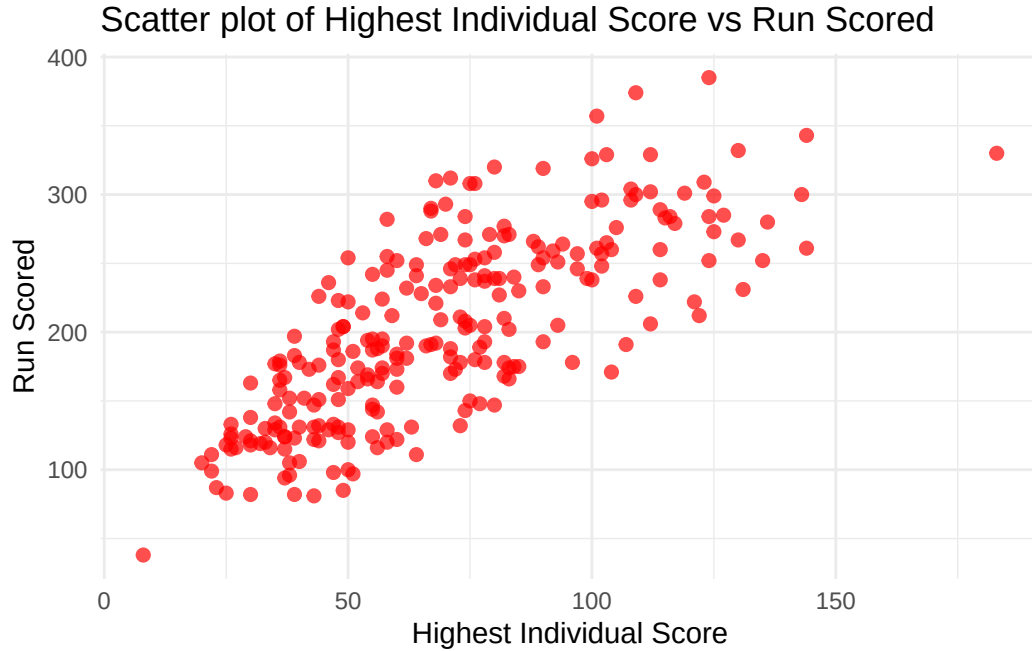


Figure 3: Relation between Highest Individual Score and Total Team Score in Asia Cup matches

Wickham 2016; Hadley Wickham et al. 2023). Models assume the match outcomes are independent.

A main limitation of this approach is that only one predictor is included per model; additional match factors that may affect total runs (e.g., team strength, opposition quality) are not considered here and could introduce omitted variable bias. These limitations are discussed further in the Discussion section.

4 Results

We report the parameter estimates and model summaries for both regression analyses:

1. Regression of Total Runs on Extras Scored

The model fit and the estimated coefficients are:

Table 1: Estimated coefficients from the regression of total runs on extras scored.

	Parameter	Estimate	Standard.error	t.value	p.value
(Intercept)	β_0 (Intercept)	177.07	8.34	21.24	<0.001
Extras.Scored	β_1 (Slope)	1.88	0.55	3.44	<0.001

Findings from the regression analysis exploring the relation between extras scored and total runs scored by the team in Asia Cup matches. The fitted linear model summarizes how changes in extras may result in change in team's total runs. Regression outputs includes an intercept $\hat{\beta}_0 = 177.07$, which indicates that a team is expected to score 177 runs in a match with 0 extras scored. This has relatively small standard error (8.34), is statistically significant and gives confidence in the predictions.

The slope coefficients $\hat{\beta}_1 = 1.88$, suggests that each additional extra run scored expects around a 2 runs increase in teams total score. This relation is statistically significant at the 5% level, as the p-value falls below the 0.05 for a t-test with null hypothesis (H_0) assumes that extras have no impact on total runs ($\beta_1 = 0$) while H_A states that the extras scored do affect total runs ($\beta_1 \neq 0$). The p-value, measures the probability of observing a coefficient as extreme as the estimated if null hypothesis were true, a low value provides strong evidence against H_0 .

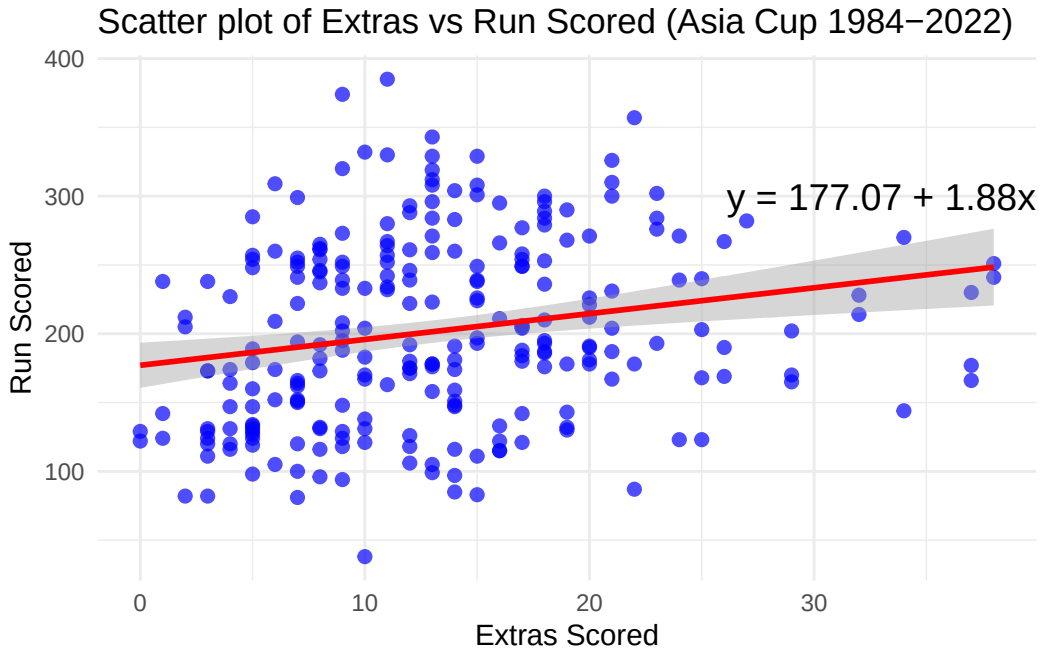


Figure 4: Scatter plot illustrating the relationship between extras scored and total runs in Asia Cup matches, with a fitted regression

Hence, the regression suggests that minimizing extras through disciplined bowling and fielding can result in a lower opponent team 's score. As conceding additional extras would increase the opponent's team total.

The Figure 4 presents the fitted regression line along with its confidence bands (shaded regions) for the relation between extras scored and total runs. The red line is the predicted values from the regression model, and the surrounding grey band shows the confidence interval around these estimates, i.e, the degree of uncertainty for the predictions. The width of this confidence band increase as the number of extras scored increase, suggesting that there will be a greater uncertainty in predicting the total score when the extras scored are more. This figure shows a support for positive association between the Extras Scored and Total Runs Scored.

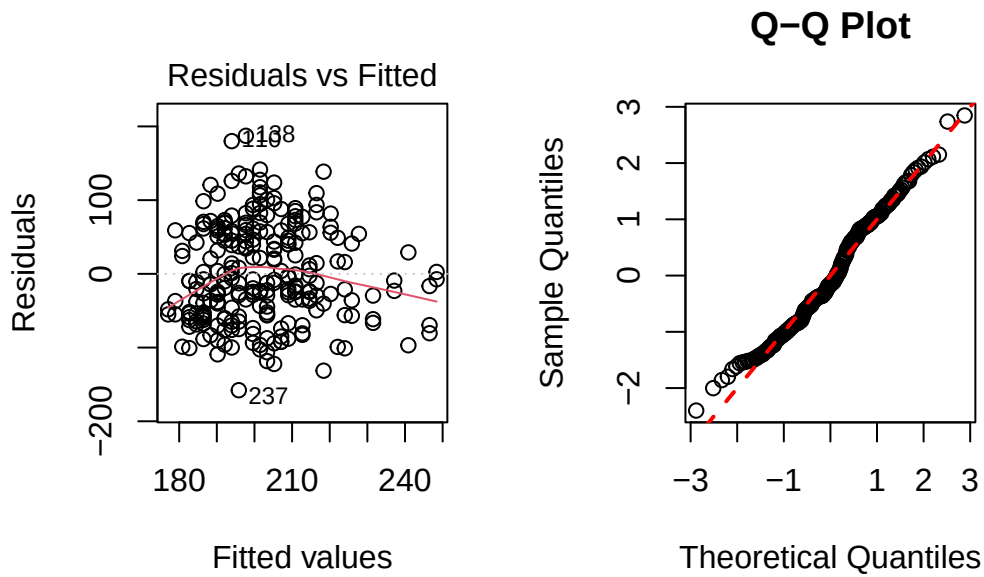


Figure 5: Regression model diagnostic plots for Asia Cup data for Run Scored regressed on Extras Scored: the left panel displays residuals versus fitted values, highlighting the spread and potential patterns; the right panel presents a Q-Q plot of standardized residuals against theoretical normal quantiles

The diagnostic plots from Figure 5 shows that the regression model is appropriate for analyzing the relation between extras and total runs in Asia Cup Cricket matches. The residuals are approximately even in distribution around zero with no strong patterns. The Q-Q plot shows a distribution that reasonably aligns well with normal distribution, indicating that the linear model's assumptions are valid with slight fluctuations at the extremes which are explainable due to the presence of outliers.

2. Regression of Total Runs on Highest Individual Score

The model fit and the estimated coefficients are:

Table 2: Estimated coefficients from the regression of total runs on highest individual score.

	Parameter	Estimate	Standard.error	t.value	p.value
(Intercept)	β_0 (Intercept)	83.20	7.02	11.85	<0.001
Highest.Score	β_1 (Slope)	1.71	0.09	18.40	<0.001

The regression analyses the relation between a team's highest individual score and its total runs gives us several insights. The intercept estimate $\hat{\beta}_0 = 83.20$ suggests that, if no player scored any runs, the team total would be predicted as 83.2. This is not directly meaningful for cricket matches, but represents a mathematical fit based on the available data. On the other hand, the slope $\hat{\beta}_1 = 1.71$ indicates a positive association, i.e, for every extra run scored by the top batsman, the team's total is expected to increase by around 1.71 runs. Both coefficients have low standard errors, which provides support to the predicted values.

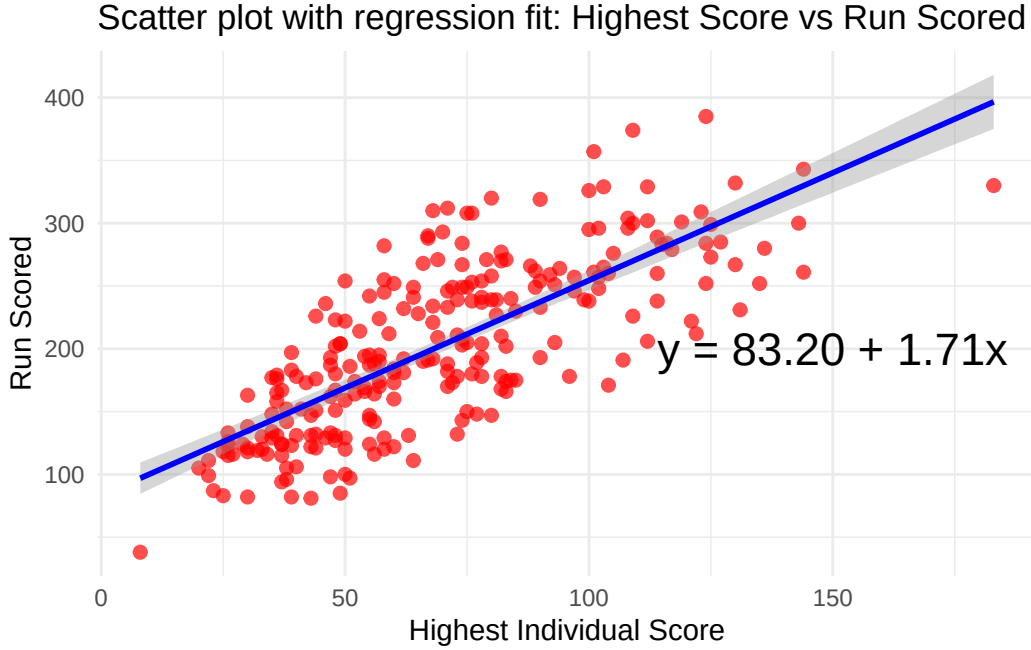


Figure 6: Scatter plot illustrating the relationship between the highest individual score and total runs scored in Asia Cup matches

The p-values being small indicate high statistical significance for a test of association between highest individual score and the team's total runs. A t-test on slope coefficient (β_1) where ($H_0 : \beta_1 = 0$), meaning that the highest individual score has no effect on team's total score, v/s ($H_A : \beta_1 \neq 0$), indicating an effect exists. p-values accompanying the slope measures the

probability of observing such a strong relation when null-hypothesis is true. Small p-value is the strong evidence against Null Hypothesis. Thus, the regression results reinforce the practical impact of top individual performances in cricket outcomes.

The scatter plot in Figure 6 with a regression line gives a summary, showing that total runs increase as the highest individual score rises. The Highest Individual Score vs Run Scored show considerable spread and a positive association, and regression line fits well to the existing data. The confidence bands around the line becomes wider for high scores, which shows increasing uncertainty while predicting total runs. The visualization confirms that the strong influence of a team's top scorer on the team's total score.

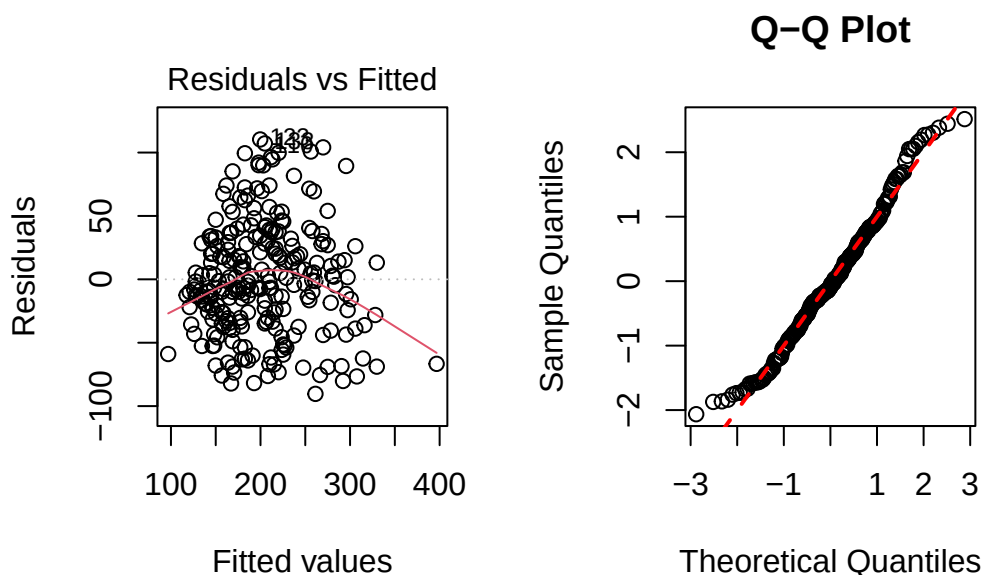


Figure 7: Regression model diagnostic plots for Asia Cup data for Runs Scored regressed on Highest Individual Score: the left panel displays residuals versus fitted values, highlighting the spread and potential patterns; the right panel presents a Q-Q plot of standardized residuals against theoretical normal quantiles

The left pane - residuals vs fitted values plot represents how much the observed total runs differ from values predicted by the regression model. The random scatter of these residuals around the origin shows that the model learns the patterns without any biases. But a downward trend at higher fitted values might be due to any outliers in the dataset. On the other hand the Q-Q plot evaluates if the residuals are distributed normally. As most of the points fall close to the normal assumption (on the $x=y$ line), this suggests that the normality assumption is largely reasonable. The non-alignment at the upper and lower ends suggests a deviation from normality at extreme scores.

Overall, these plots support the usage of linear models for this analysis, but also highlight the areas like extreme scores (high and low) where the predictions may be less accurate.

5 Discussion

This study used simple linear regressions to measure and quantify how extras scored and a team's highest individual batting score each affect total runs in Asia Cup matches. Results show that while extras provide a statistically significant but modest boost to team totals. In contrast the highest individual score has a stronger and more consistent influence on the total score of a team in the matches. These findings show that the teams with high score achievers have an improved chances to score more and thus a better chance at victory while minimizing the number of extras they concede.

Key limitations include the restricted dataset (Asia Cup matches only) and the absence of other match variables or tournament contexts. Along with the other key aspects like pitch condition, moisture in the air, team strengths have not been considered. Our linear approach may not capture all complexities of cricket scoring.

Future work should incorporate data from non-Asian teams, test additional predictors, and explore non-linear or interaction effects. With other data points like the pitch, weather, partnerships would tend to give more insights on the data.

In conclusion, the evidences suggest that exceptional individual batters who can produce high scores would yield more benefits to the teams scores than focusing exclusively on reducing extras. These insights have practical impacts while forming strategies and coaching any team for the tournaments.

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